

The third, and potentially most powerful method, is Godunov's approach. Here one gives up the simple linearization inherent in finite differencing based on Taylor series and includes the nonlinearity of the equations explicitly. There is an analytic solution for the evolution of two uniform states of a fluid separated by a discontinuity, the Riemann shock problem. Godunov's idea was to approximate the fluid by a large number of cells of uniform states, and piece them together using the Riemann solution. There have been many generalizations of Godunov's approach, which are now called *high resolution shock capturing methods*. The most influential such algorithm has probably been the PPM method [7]. General discussions of high resolution shock capturing methods and other modern algorithms are given in [8-10].

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20.2 Diffusive Initial Value Problems

Recall the model parabolic equation, the diffusion equation in one space dimension,

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial u}{\partial x} \right) \quad (20.2.1)$$

where D is the diffusion coefficient. Actually, this equation is a flux-conservative equation of the form considered in the previous section, with

$$F = -D \frac{\partial u}{\partial x} \quad (20.2.2)$$

the flux in the x -direction. We will assume $D \geq 0$, otherwise equation (20.2.1) has physically unstable solutions: A small disturbance evolves to become more and more concentrated instead of dispersing. (Don't make the mistake of trying to find a stable differencing scheme for a problem whose underlying PDEs are themselves unstable!)

Even though (20.2.1) is of the form already considered, it is useful to consider it as a model in its own right. The particular form of flux (20.2.2), and its direct generalizations, occur quite frequently in practice. Moreover, we have already seen that numerical viscosity and artificial viscosity can introduce diffusive pieces like the right-hand side of (20.2.1) in many other situations.

Consider first the case when D is a constant. Then the equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} \quad (20.2.3)$$

can be differenced in the obvious way:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D \left[\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} \right] \quad (20.2.4)$$

This is the FTCS scheme again, except that it is a second derivative that has been differenced on the right-hand side. But this makes a world of difference! The FTCS scheme was unstable for the hyperbolic equation; however, a quick calculation shows that the amplification factor for equation (20.2.4) is

$$\xi = 1 - \frac{4D\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right) \quad (20.2.5)$$

The requirement $|\xi| \leq 1$ leads to the stability criterion

$$\frac{2D\Delta t}{(\Delta x)^2} \leq 1 \quad (20.2.6)$$

The physical interpretation of the restriction (20.2.6) is that the maximum allowed timestep is, up to a numerical factor, the diffusion time across a cell of width Δx .

More generally, the diffusion time τ across a spatial scale of size λ is of order

$$\tau \sim \frac{\lambda^2}{D} \quad (20.2.7)$$

Usually we are interested in modeling accurately the evolution of features with spatial scales $\lambda \gg \Delta x$. If we are limited to timesteps satisfying (20.2.6), we will need to evolve through of order $\lambda^2/(\Delta x)^2$ steps before things start to happen on the scale of interest. This number of steps is usually prohibitive. We must therefore find a stable way of taking timesteps comparable to, or perhaps — for accuracy — somewhat smaller than, the time scale of (20.2.7).

This goal poses an immediate “philosophical” question. Obviously the large timesteps that we propose to take are going to be woefully inaccurate for the small scales that we have decided not to be interested in. We want those scales to do something stable, “innocuous,” and perhaps not too physically unreasonable. We

want to build this innocuous behavior into our differencing scheme. What should it be?

There are two different answers, each of which has its pros and cons. The first answer is to seek a differencing scheme that drives small-scale features to their *equilibrium* forms, e.g., satisfying equation (20.2.3) with the left-hand side set to zero. This answer generally makes the best physical sense; but, as we will see, it leads to a differencing scheme (“fully implicit”) that is only *first-order* accurate in time for the scales that we are interested in. The second answer is to let small-scale features *maintain* their initial amplitudes, so that the evolution of the larger-scale features of interest takes place superposed with a kind of “frozen in” (though fluctuating) background of small-scale stuff. This answer gives a differencing scheme (Crank-Nicolson) that is *second-order* accurate in time. Toward the end of an evolution calculation, however, one might want to switch over to some steps of the other kind, to drive the small-scale stuff into equilibrium. Let us now see where these distinct differencing schemes come from.

Consider the following differencing of (20.2.3):

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D \left[\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{(\Delta x)^2} \right] \quad (20.2.8)$$

This is exactly like the FTCS scheme (20.2.4), except that the spatial derivatives on the right-hand side are evaluated at timestep $n + 1$. Schemes with this character are called *fully implicit* or *backward time*, by contrast with FTCS (which is called *fully explicit*). To solve equation (20.2.8), one has to solve a set of simultaneous linear equations at each timestep for the u_j^{n+1} . Fortunately, this is a simple problem because the system is tridiagonal: Just group the terms in equation (20.2.8) appropriately:

$$-\alpha u_{j-1}^{n+1} + (1 + 2\alpha)u_j^{n+1} - \alpha u_{j+1}^{n+1} = u_j^n, \quad j = 1, 2, \dots, J-1 \quad (20.2.9)$$

where

$$\alpha \equiv \frac{D\Delta t}{(\Delta x)^2} \quad (20.2.10)$$

Supplemented by Dirichlet or Neumann boundary conditions at $j = 0$ and $j = J$, equation (20.2.9) is clearly a tridiagonal system, which can easily be solved at each timestep by the method of §2.4.

What is the behavior of (20.2.8) for very large timesteps? The answer is seen most clearly in (20.2.9), in the limit $\alpha \rightarrow \infty$ ($\Delta t \rightarrow \infty$). Dividing by α , we see that the difference equations are just the finite difference form of the equilibrium equation

$$\frac{\partial^2 u}{\partial x^2} = 0 \quad (20.2.11)$$

What about stability? The amplification factor for equation (20.2.8) is

$$\xi = \frac{1}{1 + 4\alpha \sin^2 \left(\frac{k\Delta x}{2} \right)} \quad (20.2.12)$$

Clearly $|\xi| < 1$ for any stepsize Δt . The scheme is unconditionally stable. The details of the small-scale evolution from the initial conditions are obviously inaccurate

for large Δt . But, as advertised, the correct equilibrium solution is obtained. This is the characteristic feature of implicit methods.

Here, on the other hand, is how one gets to the second of our above philosophical answers, combining the stability of an implicit method with the accuracy of a method that is second order in both space and time. Simply form the average of the explicit and implicit FTCS schemes:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{D}{2} \left[\frac{(u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}) + (u_{j+1}^n - 2u_j^n + u_{j-1}^n)}{(\Delta x)^2} \right] \quad (20.2.13)$$

Here both the left- and right-hand sides are centered at timestep $n + \frac{1}{2}$, so the method is second-order accurate in time as claimed. The amplification factor is

$$\xi = \frac{1 - 2\alpha \sin^2\left(\frac{k\Delta x}{2}\right)}{1 + 2\alpha \sin^2\left(\frac{k\Delta x}{2}\right)} \quad (20.2.14)$$

so the method is stable for any size Δt . This scheme is called the *Crank-Nicolson* scheme and is our recommended method for any simple diffusion problem (perhaps supplemented by a few fully implicit steps at the end). (See Figure 20.2.1.)

Now turn to some generalizations of the simple diffusion equation (20.2.3). Suppose first that the diffusion coefficient D is not constant, say $D = D(x)$. We can adopt either of two strategies. First, we can make an analytic change of variable

$$y = \int \frac{dx}{D(x)} \quad (20.2.15)$$

Then

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} D(x) \frac{\partial u}{\partial x} \quad (20.2.16)$$

becomes

$$\frac{\partial u}{\partial t} = \frac{1}{D(y)} \frac{\partial^2 u}{\partial y^2} \quad (20.2.17)$$

and we evaluate D at the appropriate y_j . Heuristically, the stability criterion (20.2.6) in an explicit scheme becomes

$$\Delta t \leq \min_j \left[\frac{(\Delta y)^2}{2D_j^{-1}} \right] \quad (20.2.18)$$

Note that constant spacing Δy in y does not imply constant spacing in x .

An alternative method that does not require analytically tractable forms for D is simply to difference equation (20.2.16) as it stands, centering everything appropriately. Thus the FTCS method becomes

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{D_{j+1/2}(u_{j+1}^n - u_j^n) - D_{j-1/2}(u_j^n - u_{j-1}^n)}{(\Delta x)^2} \quad (20.2.19)$$

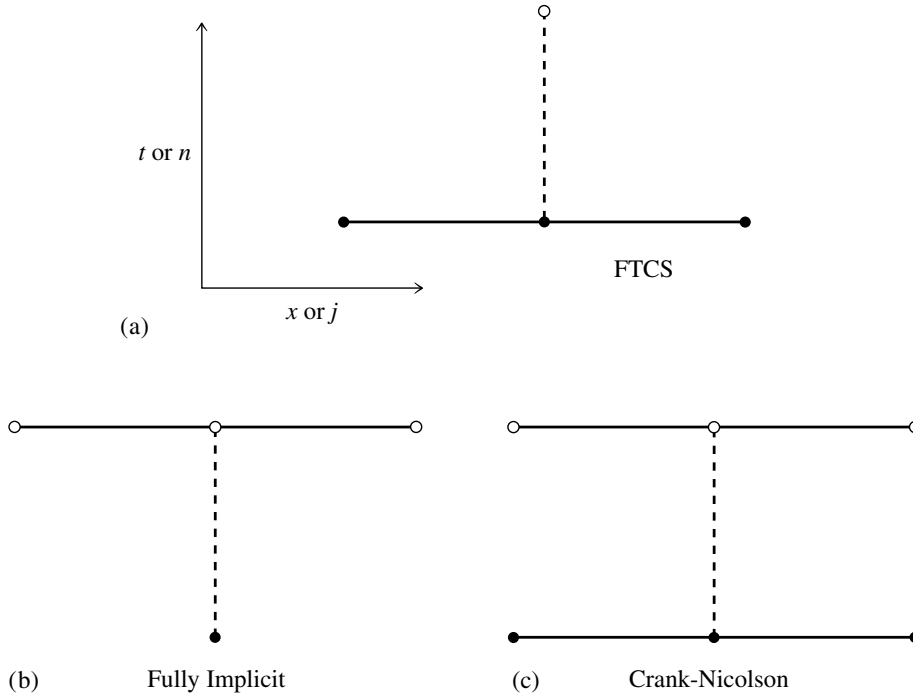


Figure 20.2.1. Three differencing schemes for diffusive problems (shown as in Figure 20.1.2). (a) Forward time centered space is first-order accurate but is stable only for sufficiently small timesteps. (b) Fully implicit is stable for arbitrarily large timesteps but is still only first-order accurate. (c) Crank-Nicolson is second-order accurate and is usually stable for large timesteps.

where

$$D_{j+1/2} \equiv D(x_{j+1/2}) \quad (20.2.20)$$

and the heuristic stability criterion is

$$\Delta t \leq \min_j \left[\frac{(\Delta x)^2}{2D_{j+1/2}} \right] \quad (20.2.21)$$

The Crank-Nicolson method can be generalized similarly.

The second complication one can consider is a nonlinear diffusion problem, for example where $D = D(u)$. Explicit schemes can be generalized in the obvious way. For example, in equation (20.2.19) write

$$D_{j+1/2} = \frac{1}{2} [D(u_{j+1}^n) + D(u_j^n)] \quad (20.2.22)$$

Implicit schemes are not as easy. The replacement (20.2.22) with $n \rightarrow n + 1$ leaves us with a nasty set of coupled nonlinear equations to solve at each timestep. Often there is an easier way: If the form of $D(u)$ allows us to integrate

$$dz = D(u)du \quad (20.2.23)$$

analytically for $z(u)$, then the right-hand side of (20.2.1) becomes $\partial^2 z / \partial x^2$, which

we difference implicitly as

$$\frac{z_{j+1}^{n+1} - 2z_j^{n+1} + z_{j-1}^{n+1}}{(\Delta x)^2} \quad (20.2.24)$$

Now linearize each term on the right-hand side of equation (20.2.24), for example

$$\begin{aligned} z_j^{n+1} &\equiv z(u_j^{n+1}) = z(u_j^n) + (u_j^{n+1} - u_j^n) \left. \frac{\partial z}{\partial u} \right|_{j,n} \\ &= z(u_j^n) + (u_j^{n+1} - u_j^n) D(u_j^n) \end{aligned} \quad (20.2.25)$$

This reduces the problem to tridiagonal form again and in practice usually retains the stability advantages of fully implicit differencing.

20.2.1 Schrödinger Equation

Sometimes the physical problem being solved imposes constraints on the differencing scheme that we have not yet taken into account. For example, consider the time-dependent Schrödinger equation of quantum mechanics. This is basically a parabolic equation for the evolution of a complex quantity ψ . For the scattering of a wavepacket by a one-dimensional potential $V(x)$, the equation has the form

$$i \frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + V(x)\psi \quad (20.2.26)$$

(Here we have chosen units so that Planck's constant $\hbar = 1$ and the particle mass $m = 1/2$.) One is given the initial wavepacket, $\psi(x, t = 0)$, together with boundary conditions that $\psi \rightarrow 0$ at $x \rightarrow \pm\infty$. Suppose we content ourselves with first-order accuracy in time but want to use an implicit scheme, for stability. A slight generalization of (20.2.8) leads to

$$i \left[\frac{\psi_j^{n+1} - \psi_j^n}{\Delta t} \right] = - \left[\frac{\psi_{j+1}^{n+1} - 2\psi_j^{n+1} + \psi_{j-1}^{n+1}}{(\Delta x)^2} \right] + V_j \psi_j^{n+1} \quad (20.2.27)$$

for which

$$\xi = \frac{1}{1 + i \left[\frac{4\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right) + V_j \Delta t \right]} \quad (20.2.28)$$

This is unconditionally stable but unfortunately is not *unitary*. The underlying physical problem requires that the total probability of finding the particle somewhere remains unity. This is represented formally by the modulus-square norm of ψ remaining unity:

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1 \quad (20.2.29)$$

The initial wave function $\psi(x, 0)$ is normalized to satisfy (20.2.29). The Schrödinger equation (20.2.26) then guarantees that this condition is satisfied at all later times.

Let us write equation (20.2.26) in the form

$$i \frac{\partial \psi}{\partial t} = H \psi \quad (20.2.30)$$